# EXPERIMENT 7

## 7.1.1 Inverted Star Pattern:

### Algorithm:

Step 1: Start

Step 2: Read an integer value n from the user.

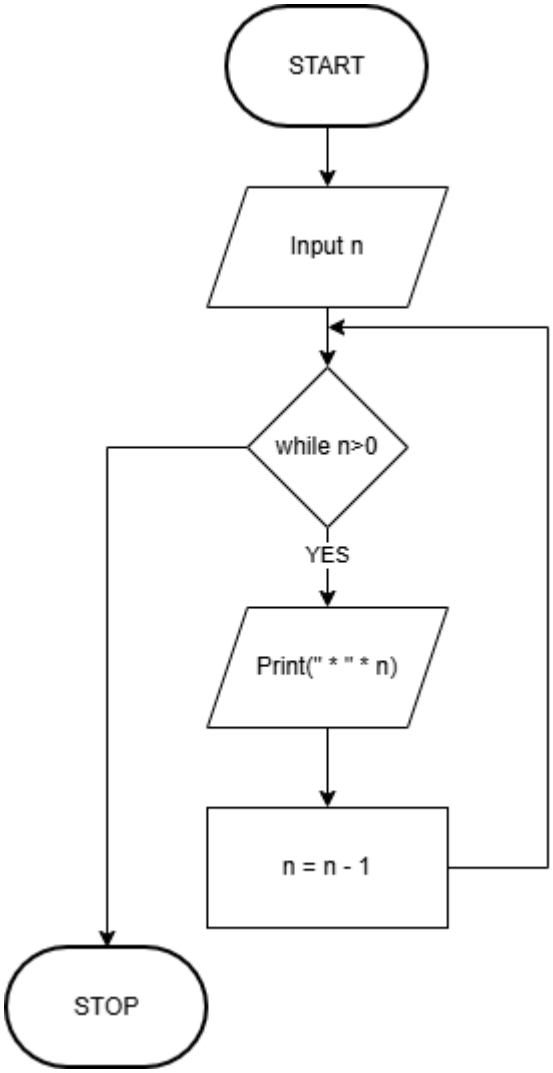
Step 3: Initialize a loop from i = n down to 1 (decreasing by 1 each time).

Step 4: For each value of i, print i number of * (stars).

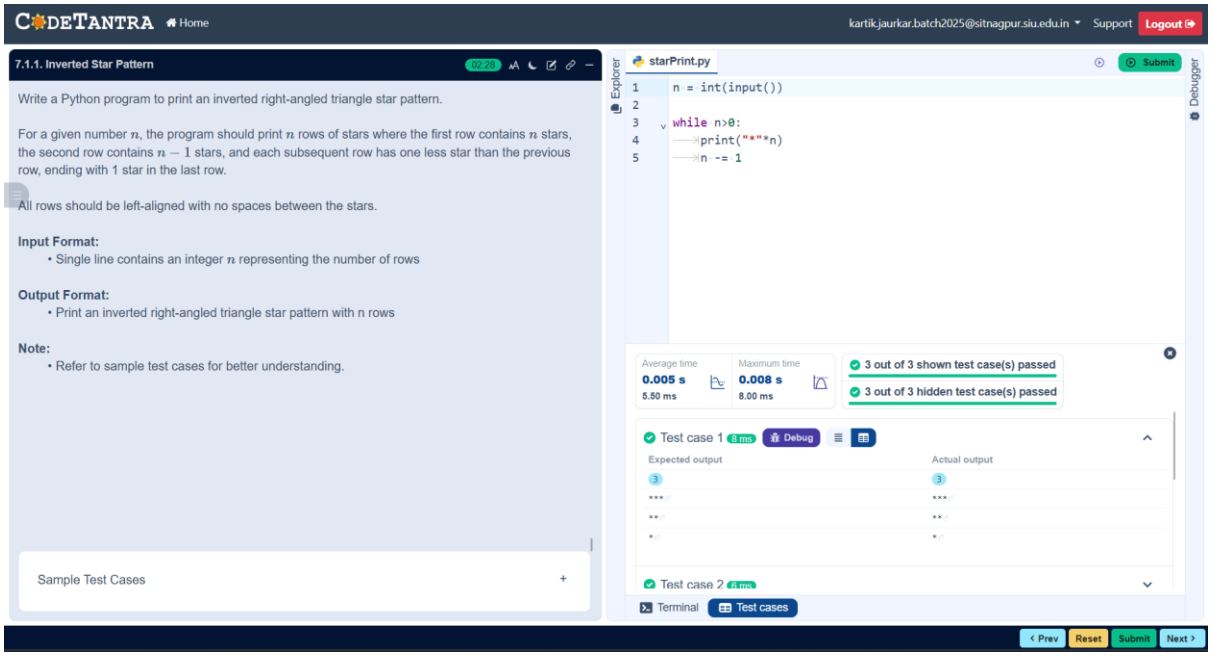
Step 5: Repeat Step 4 until the loop ends (when i becomes 0).

Step 6: Stop

## Flowchart:



# Execution: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7.1.1. Inverted Star Pattern
02:28
Write a Python program to print an inverted right-angled triangle star pattern.
For a given number $n$, the program should print $n$ rows of stars where the first row contains $n$ stars, the second row contains $n-1$ stars, and each subsequent row has one less star than the previous row, ending with 1 star in the last row.
All rows should be left-aligned with no spaces between the stars.
Input Format:
• Single line contains an integer $n$ representing the number of rows
Output Format:
• Print an inverted right-angled triangle star pattern with n rows
Note:
• Refer to sample test cases for better understanding.
Sample Test Cases
+
Explorer
starPrint.py
Submit
Debugger
n = int(input())
while n>0:
print("*"*n)
n -= 1
Average time
0.005 s
5.50 ms
Maximum time
0.008 s
8.00 ms
3 out of 3 shown test case(s) passed
3 out of 3 hidden test case(s) passed
Test case 1
8 ms
Debug
Expected output
Actual output
3
***
**
*
Test case 2
Terminal
Test cases
Prev
Reset
Submit
Next

# 7.1.2 Possible combination from 3 digits:

## Algorithm:

**Step 1:** Start

**Step 2:** Take three inputs a, b, and c from the user separated by spaces.

**Step 3:** Store the inputs in a list called digits.

**Step 4:** Use three nested loops to generate index values:

- Loop i from 0 to 2
- Loop j from 0 to 2
- Loop k from 0 to 2

**Step 5:** Inside the loops, check the condition:

- If i, j, and k are all different ($i \neq j$, $i \neq k$, $j \neq k$)

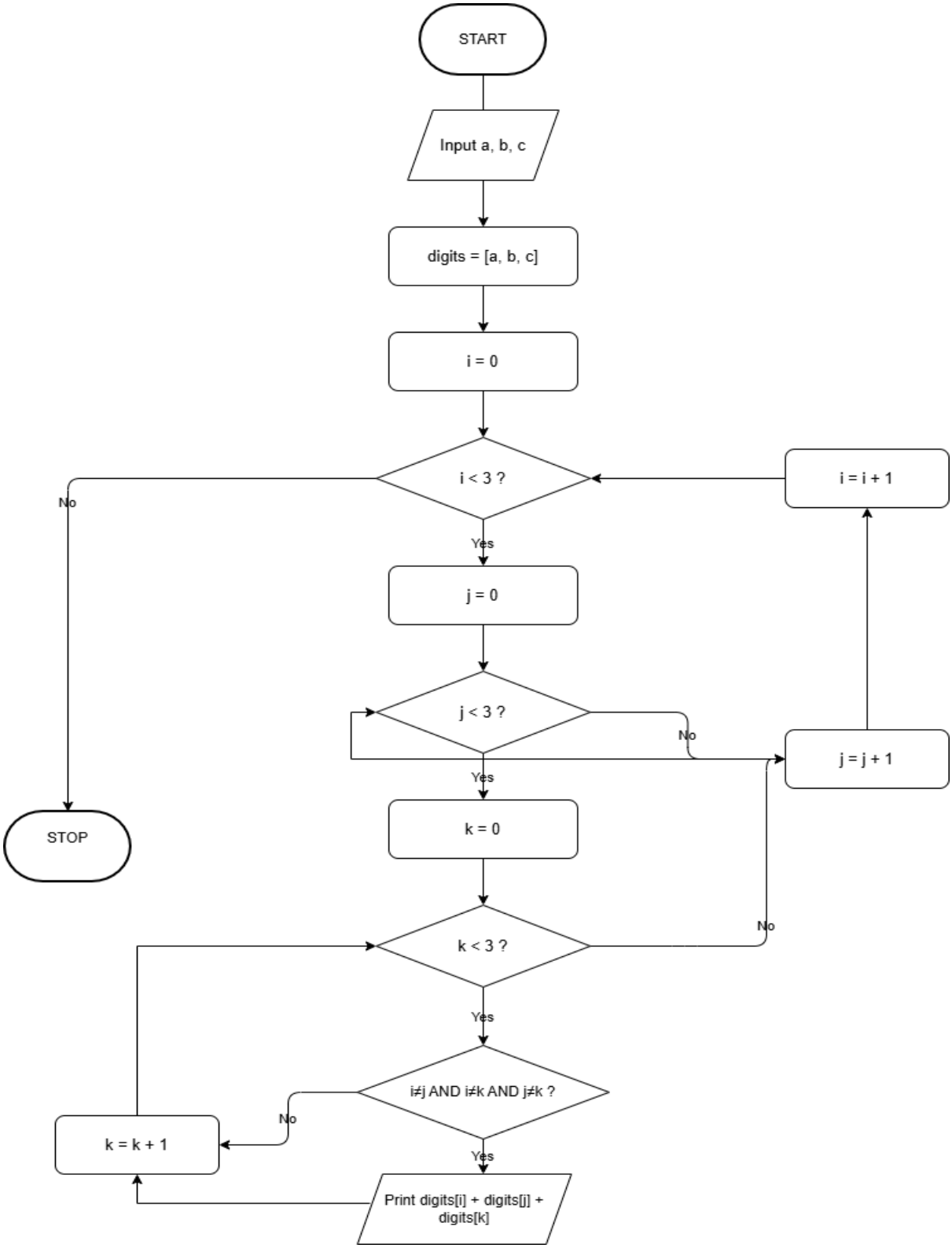
**Step 6:** If the condition is true:

- Concatenate digits[i], digits[j], and digits[k]
- Print the result

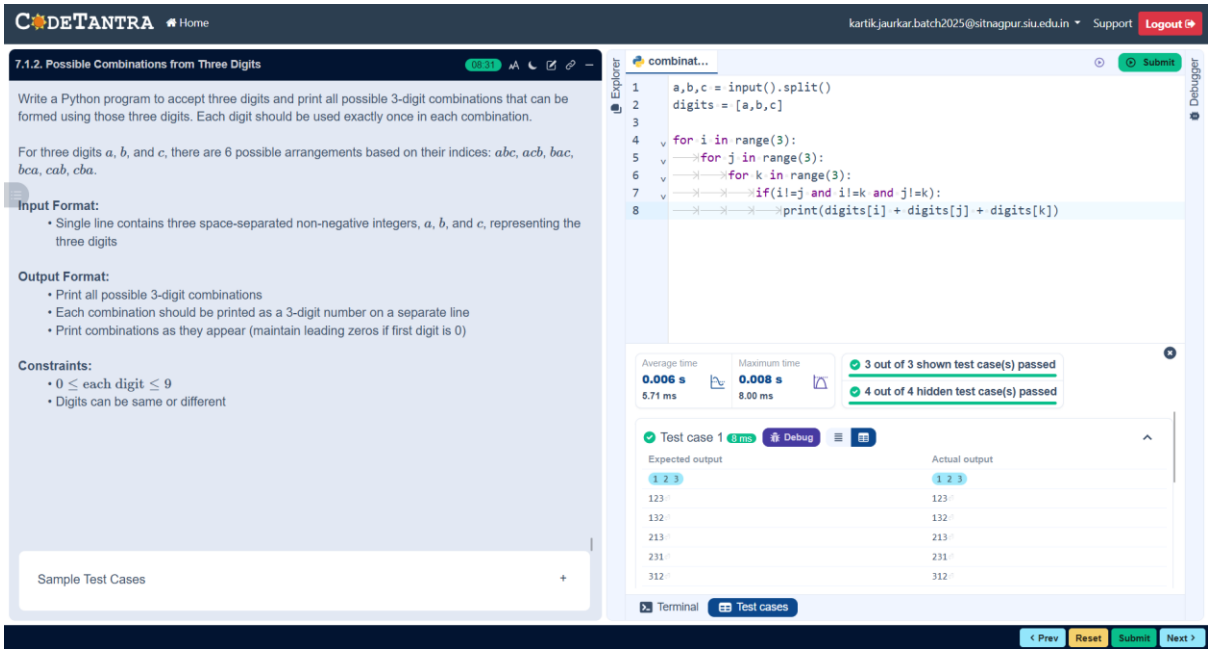
**Step 7:** Continue the loops until all combinations are checked.

**Step 8:** Stop

# Flowchart


START
Input a, b, c
digits = [a, b, c]
i = 0
i < 3 ?
No
Yes
i = i + 1
j = 0
j < 3 ?
No
Yes
j = j + 1
STOP
k = 0
k < 3 ?
No
Yes
i≠j AND i≠k AND j≠k ?
No
Yes
k = k + 1
Print digits[i] + digits[j] + digits[k]

# Execution: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7.1.2. Possible Combinations from Three Digits
08:31
Write a Python program to accept three digits and print all possible 3-digit combinations that can be formed using those three digits. Each digit should be used exactly once in each combination.
For three digits $a$, $b$, and $c$, there are 6 possible arrangements based on their indices: $abc$, $acb$, $bac$, $bca$, $cab$, $cba$.
Input Format:
• Single line contains three space-separated non-negative integers, $a$, $b$, and $c$, representing the three digits
Output Format:
• Print all possible 3-digit combinations
• Each combination should be printed as a 3-digit number on a separate line
• Print combinations as they appear (maintain leading zeros if first digit is 0)
Constraints:
• $0 \leq \text{each digit} \leq 9$
• Digits can be same or different
Sample Test Cases
Explorer
combinat...
Submit
Debugger
a,b,c = input().split()
digits = [a,b,c]
for i in range(3):
for j in range(3):
for k in range(3):
if(i!=j and i!=k and j!=k):
print(digits[i] + digits[j] + digits[k])
Average time
0.006 s
5.71 ms
Maximum time
0.008 s
8.00 ms
3 out of 3 shown test case(s) passed
4 out of 4 hidden test case(s) passed
Test case 1
8 ms
Debug
Expected output
Actual output
1 2 3
123
132
213
231
312
Terminal
Test cases
Prev
Reset
Submit
Next